Data Extraction in ETL

Assignment Questions

Question 1: Describe different types of data sources used in ETL with suitable examples.

ANSWER:

Excel sheets, CSV, API, SQL

- Customer database – MYSQL
- Sales database – CSV
- Website trafficking - API

Database = (oracle, MYSQL), NONSQL(MANGODB)

Cloud storage = Cloud sources: Google Sheets, AWS S3, Azure Blob for storing large files or logs.

File-based sources: Excel and CSV files

Each source contributes different formats and structures, making ETL necessary to combine them

Question 2: What is data extraction? Explain its role in the ETL pipeline.

ANSWER:

In data extraction extract the raw data and collected in its original form before cleaning or processing.

Create database in MYSQL, NONSQL, PAYTHON, POWERBI, EXCEL

Data extraction is the process of retrieving data from various locations such as files, databases, and APIs.

Create link between ETL and database

Etl first step pipeline

Without extraction transformation and load cannot happened. data will be collect multiple sources.

Question 3: Explain the difference between CSV and Excel in terms of extraction and ETL usage.

ANSWER:

Csv

- Csv file is light and faster.
- Csv can do automation, export file, large dataset
- Csv file can be extracted using Excel, PAYTHON, POWERBI.
- Csv high ETL usage.

Excel

- Excel heavy database, charts, formula, multiple sheets.
- Excel can be manually prepared and sharing between the team.
- Excel can be extracted using excel, PAYTHON, power query.
- Excel small scale business use ETL.

Question 4 : Explain the steps involved in extracting data from a relational database.

ANSWER:

Database extraction is the process of retrieving data from

Relational database.

1. Establish Connection
2. Authentication

   The system verifies user credentials and checks permissions on tables or views
3. Query Execution

   Querying: SQL commands like SELECT, JOIN, WHERE are executed to fetch relevant data.
4. Fetch Results (rows returned).
5. Export Data

   The result is exported as CSV/Excel or directly loaded into the ETL pipeline for transformation.
6. Handle Errors (timeouts, missing records).

Question 5: Explain three common challenges faced during data extraction.

ANSWER:

- Large files
- Authentication issues
- Data quality changes

- Schema changes

Inconsistent formats: Different date formats (DD/MM/YYYY vs. MM-DD-YY) cause mismatches, making merging difficult.

API rate limits: APIs restrict the number of requests, causing failures if the ETL runs too frequently.

Question 6: What are APIs? Explain how APIs help in real-time data extraction.

ANSWER:

Api allowed the communicate and data exchange process.

Use weather API, Stock API, Social media API.

Api enable real time data pull from external system.

Api real time data updated frequently without manual support.

Question 7: Why are databases preferred for enterprise-level data extraction?

ANSWER:

- Structure & reliable
- Scalability
- Security
- Integration

Enterprises prefer database they ensure consistency, performance.

They can handle millions of records and support indexing for fast retrieval.

User authentication and permission controls ensure secure access.

SQL provides powerful querying to extract exactly the required data without loading everything

Question 8: What steps should an ETL developer take when extracting data from large CSV files (1GB+)?

ANSWER:

- Schema inconsistencies
- Poor quality data
- Large size files
- Compression
- Parrel processing
- Unhandled delimiters
- Use optimized libraries (e.g., Dask or PySpark) for distributed processing.[SEP]These steps ensure smooth extraction without system crash.